

Istanbul Technical University Database Systems Project Report

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1 Introduction

The objective of this project was to design and develop a web-based application to visualize and analyze World Development Indicators (WDI).

The motivation behind this project stems from the issue of **Data Fragmentation**, where critical global data is scattered across disparate sources (World Bank, WHO, UNESCO), making cross-domain analysis difficult. Furthermore, traditional flat-file storage often leads to massive redundancy and update anomalies.

Our goal was to centralize this information into a normalized (3NF) relational schema, implement a **Strict Admin Approval System** for security, and transform raw static data into dynamic, comparative time-series visualizations.

2 Tools and Technologies

The project was built using a robust stack of open-source technologies:

2.1 Relational Database Management System (RDBMS)

MySQL was chosen as the RDBMS for this project.

- **Reason for choice:** MySQL offers excellent performance for read-heavy operations, which is essential for a data visualization platform. It supports the necessary foreign key constraints to handle the relationships between Countries, Regions, and Indicators.

2.2 Programming Language and Framework

Python was selected as the core programming language, utilizing the **Flask** web framework.

- **Flask:** A lightweight WSGI web application framework used for routing and back-end logic.
- **Jinja2 Templating:** Used to dynamically render HTML pages with data fetched from the backend.

2.3 Additional Tools

- **Frontend:** HTML5, CSS3, and Bootstrap for responsive design.
- **Data Visualization:** JavaScript libraries were used to render interactive charts for the comparison modules.
- **Version Control:** Git and GitHub.

3 Database Design

3.1 Entity-Relationship (ER) Diagram

The database follows a normalized relational schema designed to reduce redundancy. The design utilizes an **ISA (Is-A) relationship** hierarchy to categorize specific indicator types (Health, Economy, Education) while maintaining a central definition.

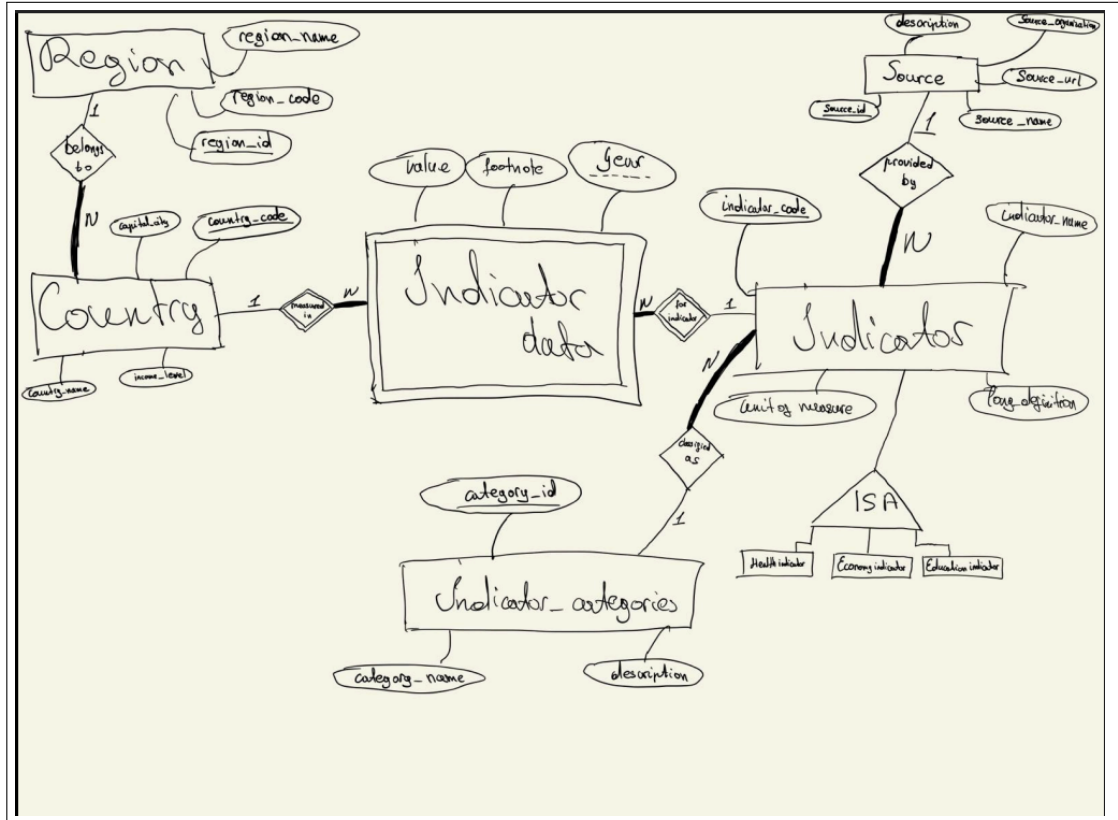


Figure 1: Entity Relationship Diagram of the WDI Database

3.2 Schema Explanation

The database consists of several key tables designed to achieve 3rd Normal Form (3NF) compliance:

1. **Regions & Countries:** Stores static nation data. The **Countries** table is linked to **Regions** via a foreign key.
2. **Sources:** Stores organization details (e.g., "World Bank"). By separating this into its own table, we eliminated the redundancy of repeating source names for thousands of rows.
3. **IndicatorCategories:** Categorizes data into broader topics to facilitate browsing.
4. **Indicators:** The central definition table. It links to categories and sources.
5. **ISA Hierarchy (Specialized Tables):** To optimize topic-specific queries, we implemented specialized tables: **HealthIndicators**, **EconomyIndicators**, and **EducationIndicators**.

These inherit structure from the main definition but allow for domain-specific categorization.

6. **IndicatorData:** The primary fact table linking **Countries** and **Indicators** with **Year** and **Value**.

3.3 Integrity and Delete Constraints

To maintain database integrity, specific ‘ON DELETE’ constraints were implemented:

- **IndicatorData:** ON DELETE RESTRICT for **Countries** and **Indicators** (preventing accidental loss of historical data).
- **Specialized Tables (e.g., HealthIndicators):** ON DELETE CASCADE for **IndicatorCategories**, ensuring that if a category is removed, the associated specialized links are cleaned up.
- **Countries:** ON DELETE RESTRICT for **Regions**.

4 Dataset Outline

The dataset is derived from the World Development Indicators (WDI) database managed by the World Bank.

- **Volume:** The database contains approximately **19,962 data points**.
- **Coverage:** It tracks information for **211 countries** across **14 active regions**.
- **Domains:** The data covers three core domains: Economy, Health, and Education.

5 Implementation Details

5.1 Database Initialization and Connection Management

The system utilizes the `mysql.connector` library to interface with the MySQL database. To ensure modularity and robustness, the connection logic is encapsulated within the `get_db_connection` function. This function manages the lifecycle of the database session by invoking the `connect()` method and is specifically designed to handle potential errors or exceptions that may occur during the connection establishment phase.

The initialization process is orchestrated by the `init_db` function, which performs the following operations:

1. **Connection Establishment:** A connection object and a cursor are retrieved using the aforementioned `get_db_connection` method.
2. **Schema Setup:** The function reads the external SQL script `database/create_users_table.sql` and executes it to ensure the required **Users** table structure exists.
3. **Admin User Configuration:** The system performs a check to see if the default administrator account (username: ‘huseynov’) already exists.

- If the user is not found, the system generates a secure password hash using the `pbkdf2:sha256` method from the `werkzeug.security` library.
 - The admin user is then inserted into the database with the hashed password and the 'admin' role.
4. **Finalization:** The transaction is committed to the database, and both the cursor and connection are closed to prevent resource leaks.

5.2 ETL and Normalization Strategy

To handle the raw JSON input from the WDI API, we developed a Python ETL (Extract, Transform, Load) script.

- **In-Memory De-duplication:** We utilized Python dictionaries ('source_map') to identify and filter unique entries (such as Sources and Categories) before database insertion. This prevented duplicate records and ensured clean foreign key associations.
- **Transformation:** Nested JSON objects were flattened and mapped to the relational schema, ensuring that source details were stored once and referenced by ID.

5.3 Complex Queries and Optimization

The application utilizes complex SQL queries involving multiple 'JOIN' operations to fetch indicators along with their sources and categories. To handle large result sets, pagination ('LIMIT' and 'OFFSET') was implemented in the backend queries.

```

1 # Logic for fetching indicators with pagination
2 sql = """
3     SELECT i.*, s.source_name, c.category_name
4     FROM EducationIndicators i
5     LEFT JOIN Sources s ON i.source_id = s.id
6     LEFT JOIN IndicatorCategories c ON i.category_id = c.id
7     LIMIT %s OFFSET %s
8 """

```

Listing 1: Example of Pagination Query logic

6 Website Features and Functionality

6.1 Interactive Dashboard

The Dashboard provides an overview of global metrics. It features "Rising Stars" (top-performing countries) and visualizations such as the **Global Population Trend** and **Global GDP Growth**.

6.2 Comparative Analysis Tool

The core analysis module allows users to:

- Select multiple countries (e.g., Argentina, Canada) to compare against a specific indicator.

- View generated line charts for trend analysis.
- **Export Data:** Users can export the comparison results to Excel for offline use.

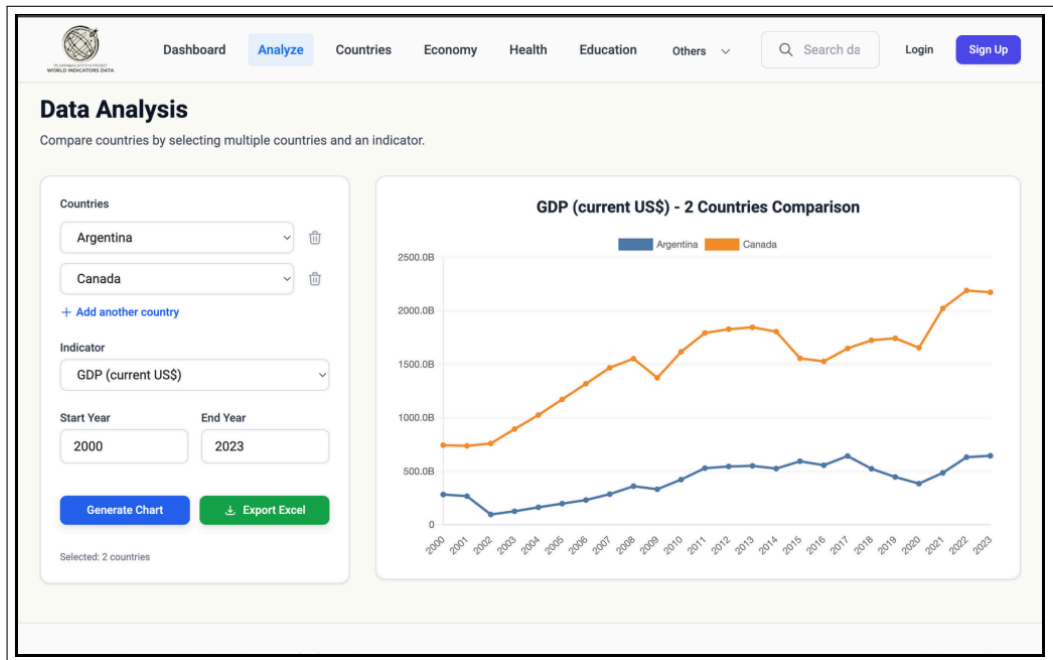


Figure 2: Country Comparison Module

6.3 Administrative Panel

The system includes a secured Admin Panel with the following features:

1. **Strict Admin Approval:** New users cannot access the system immediately; they must be explicitly approved by an Admin.
2. **CRUD Operations:** Admins can Add, Edit, or Delete indicators and countries.
3. **Validation:** Specific constraints are enforced in the UI, such as preventing the modification of a "Country Code" once created.

7 Team Contributions

The project's development was distributed to ensure coverage of all full-stack components:

Ali Huseynov:

Fronted / Backend Operations and UI Views. Specifically managed the **Countries Table**.

Ahmet Yusuf Kurukız:

Backend Operations. Responsible for the structure and management of the main **Indicators Table**.

Devrim Polat:

Backend Operations. Specifically responsible for the **Economy Indicators Table**.

Mehmet Fatih Kaya:

Frontend Operations and UI Views. Managed the ****Education Indicators Table****.

Yunus Korkmaz:

Database Management. Specifically responsible for the ****Health Indicators Table****.

8 Difficulties and Conclusion

8.1 Difficulties Encountered

- **Raw Data Redundancy:** The raw API response contained nested JSON where source details were repeated for thousands of rows. This required the custom ETL script described in Section 5 to normalize the data into 3NF.
- **API Unreachability:** During data fetching, we encountered server errors (500, 502). We implemented a retry mechanism (5 attempts) to handle these connection issues.

8.2 Conclusion

The project successfully delivered a functional web application for analyzing World Development Indicators. By transforming fragmented, flat-file data into a structured relational database, we achieved significant improvements in data consistency and query efficiency. The inclusion of a strict admin approval system and domain-specific indicator tables ensures both the security and usability of the platform.